

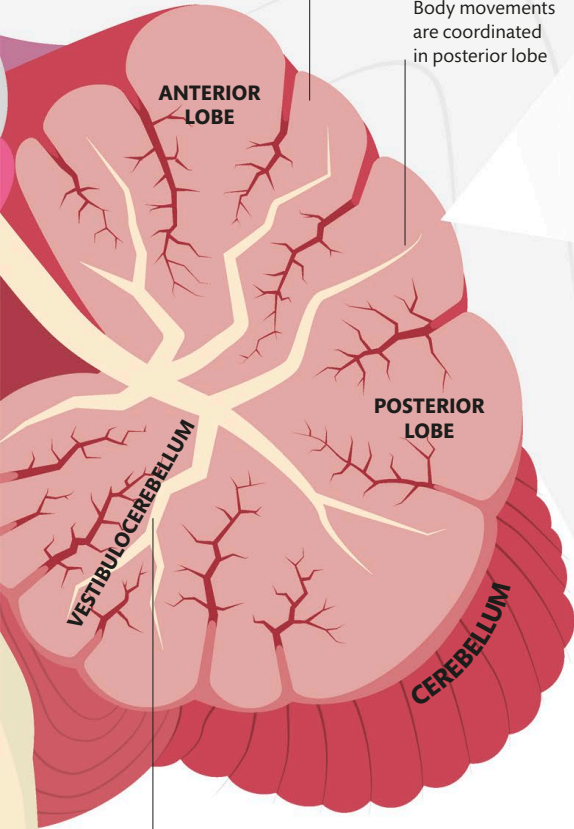


The little brain

The cerebellum, a term that means "little brain," is a highly folded region of the hindbrain that sits behind the brain stem. Like the cerebrum above it, the cerebellum is divided into two lobes. These are divided laterally into functional zones.

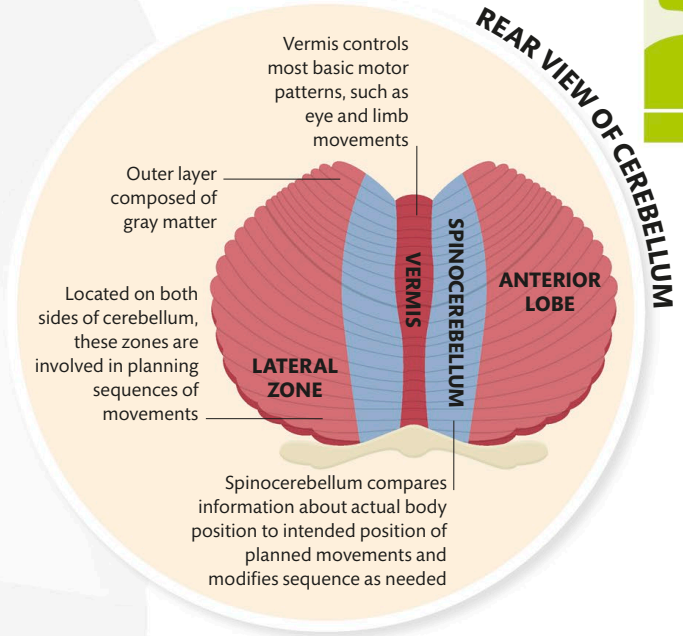
Anterior lobe of cerebellum receives information about body posture from spinal cord

Body movements are coordinated in posterior lobe



Vestibulocerebellum is involved in head control, eye movements, and maintaining balance through information from the inner ear

**KNOWLEDGE OF THE
CEREBELLUM WAS
ADVANCED BY
STUDYING BRAIN-
INJURED SOLDIERS
IN WWI**



The cerebellum

Although the cerebellum appears to play a part in maintaining attention and processing language, it is most associated with its role in the regulation of body movement. Specifically, its role is to convert the broad executive motor requests into smooth and coordinated muscle sequences, error-correcting all the while. It routes its outputs through the thalamus. At the microscopic level, the cerebellum's cells are arranged in layers. The purpose of these layers is to lay down fixed neural pathways for all kinds of learned movement patterns, such as walking, talking, and keeping balance. Damage to the cerebellum does not result in paralysis, but slow jerky movements.

THE CEREBELLUM AND NEURAL NETWORKS

Some artificial intelligences (AI) use a system inspired by the anatomy of the cerebellum. AI programs itself by machine learning. It does this with a processor called a neural network, where inputs find their way by trial and error through layers of connections, a setup that mirrors the way the cerebellum lays down patterns for learned movements.

